

Customer Retention Analysis Using BG/NBD Model

Abstract

Customer retention is a key factor in improving business growth and profitability. The Beta-Geometric/Negative Binomial Distribution (BG/NBD) model is a probabilistic model used to analyze customer purchasing behavior and predict whether customers are likely to remain active. This project applies the BG/NBD model to customer transaction data to estimate retention rates, identify customers at risk of churn, and support data-driven marketing strategies. The results help organizations improve customer loyalty and maximize long-term revenue.

Introduction

Retaining existing customers is more cost-effective than acquiring new ones. Businesses need reliable methods to understand customer behavior and predict future purchasing activity. The BG/NBD model uses historical transaction data to estimate the probability that a customer is still active and likely to make future purchases. This project focuses on customer retention analysis using the BG/NBD model.

Objectives

- Analyze customer purchasing behavior.
- Predict the probability that customers remain active.
- Identify customers who are at risk of churn.
- Improve customer retention strategies.
- Support marketing decisions using predictive analytics.

Methodology

1. Collect customer transaction data.
2. Clean and preprocess the dataset.
3. Calculate Recency, Frequency, and Customer Age (T).
4. Train the BG/NBD model.
5. Estimate the probability that each customer is still active.
6. Evaluate customer retention and generate analytical reports.

Dataset

The dataset includes:

- Customer ID
- Transaction Date
- Purchase Frequency
- Recency

- Customer Age (T)
- Transaction Value

Results

The BG/NBD model effectively identifies active customers and customers likely to stop purchasing. The analysis enables businesses to target retention campaigns toward high-value customers and reduce customer churn.

Conclusion

Customer Retention Analysis using the BG/NBD model provides valuable insights into customer behavior and future purchasing patterns. It helps businesses increase customer loyalty, optimize marketing campaigns, and improve long-term profitability through data-driven decision-making.

References

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